C

ACT SUBJECT TEST — MATH 1 • UNIT 0.01

ACT II Math Level 1 Diagnostic & Strategy Drill

ANSWER KEY & SOLUTIONS

Algebra & Geometry Readiness • Official ACT Subject Test Curriculum

Total Questions: 25 • Verified Step-by-Step Solutions

QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 C	Q2 A	Q3 B	Q4 A	Q5 B
Q6 B	Q7 B	Q8 B	Q9 C	Q10 D
Q11 B	Q12 A	Q13 D	Q14 D	Q15 C
Q16 B	Q17 D	Q18 D	Q19 B	Q20 D
Q21 D	Q22 C	Q23 A	Q24 D	Q25 A

ACT II Math Level 1 Diagnostic & Strategy Drill Practice Exercises

1 MCQ

In the standard (x, y) coordinate plane, the lines with equations $2x - y = 7$ and $x + y = 5$ intersect at the point (a, b). What is the value of a?

A 1

B 3

C 4 ✓

D 5

WORKED SOLUTION & VERIFICATION

The point of intersection (a, b) satisfies both linear equations simultaneously:

1) $2a - b = 7$

2) $a + b = 5$

To solve for a, add the two equations together to eliminate b:

$$(2a - b) + (a + b) = 7 + 5$$

$$3a = 12$$

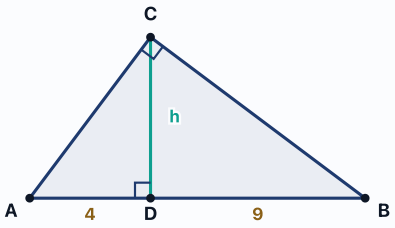
$$a = 4$$

Substituting $a = 4$ into the second equation gives $4 + b = 5$, which yields $b = 1$. Thus, the lines intersect at $(4, 1)$, and the value of a is 4.

- A common misconception is solving for $b = 1$ and selecting 1 (Choice A).
- Another error is an arithmetic mistake in dividing 12 by 3 to get 3 (Choice B).
- Choice D represents the constant from the second equation without solving the system.

2 MCQ

In right triangle ABC, altitude $\overline{BD}$ is drawn to hypotenuse $\overline{AC}$. If $AD = 4$ and $DC = 9$, what is the length of altitude $\overline{BD}$?



- **A** 6 ✓
- **B** 6.5
- **C** 13
- **D** 36

WORKED SOLUTION & VERIFICATION

By the Geometric Mean Theorem (Altitude Rule) for right triangles, the altitude to the hypotenuse is the geometric mean of the two segments into which it divides the hypotenuse:

$$BD^2 = AD \cdot DC$$

Substitute the given lengths $AD = 4$ and $DC = 9$:

$$BD^2 = 4 \cdot 9 = 36$$

Taking the positive square root:

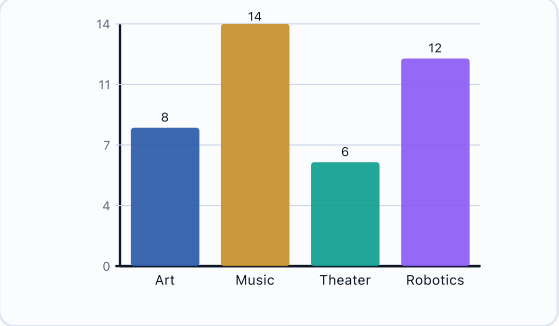
$$BD = \sqrt{36} = 6$$

Common Misconceptions:

- Choice B (6.5) confuses the geometric mean with the arithmetic mean: $\frac{4+9}{2} = 6.5$.
- Choice C (13) incorrectly adds the two segments together: $4 + 9 = 13$.
- Choice D (36) finds $BD^2 = 4 \times 9$ but forgets to take the square root.

3 MCQ

The bar graph below shows the number of students enrolled in each of four elective courses at a high school. If one student is selected at random from these four courses, what is the probability that the selected student is enrolled in Music?



A $\frac{7}{50}$

B $\frac{7}{20}$ ✓

C $\frac{7}{13}$

D $\frac{13}{20}$

WORKED SOLUTION & VERIFICATION

To find the probability that a randomly chosen student is enrolled in Music, divide the number of students enrolled in Music by the total number of students enrolled across all four courses.

1. Find the total number of students:

$$\text{Total} = 8 + 14 + 6 + 12 = 40$$

2. Find the number of students in Music:

$$n(\text{Music}) = 14$$

3. Calculate the probability:

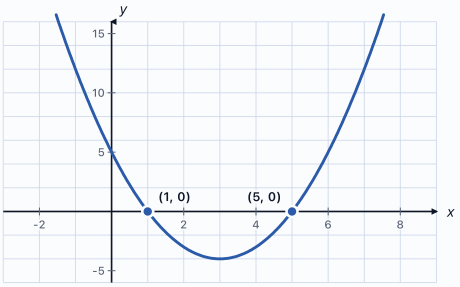
$$P(\text{Music}) = \frac{14}{40} = \frac{7}{20}$$

Common misconceptions:

- $\frac{7}{13}$ computes the odds in favor (ratio of students in Music to students not in Music, $\frac{14}{26}$) rather than probability.
- $\frac{13}{20}$ represents the complement probability (the probability that the student is NOT enrolled in Music, $1 - \frac{7}{20} = \frac{13}{20}$).
- $\frac{7}{50}$ incorrectly assumes the total is 100 students.

4 MCQ

The graph of the quadratic function $f(x) = x^2 - 6x + 5$ is shown in the standard (x, y) coordinate plane below. A horizontal line $y = k$ intersects the graph of $f(x)$ at two points that are 8 units apart. What is the value of k?



A 12 ✓

B 8

C 4

D -3

WORKED SOLUTION & VERIFICATION

We are given the quadratic function $f(x) = x^2 - 6x + 5$ and a horizontal line $y = k$ that intersects the parabola at two points separated by a horizontal distance of 8.

Method 1: Using the axis of symmetry

The axis of symmetry for a parabola in standard form $y = ax^2 + bx + c$ is given by:

$$x = -\frac{b}{2a} = -\frac{-6}{2(1)} = 3$$

Because parabolas are symmetric about their axis of symmetry, the two intersection points on the horizontal line $y = k$ must lie equidistant from $x = 3$. Since the total distance between them is 8, each point lies $\frac{8}{2} = 4$ units away from the line of symmetry:

$$x_1 = 3 - 4 = -1$$

$$x_2 = 3 + 4 = 7$$

Since both points lie on the graph of $f(x)$, the value of k is found by evaluating $f(x)$ at either $x = -1$ or $x = 7$:

$$k = f(7) = (7)^2 - 6(7) + 5 = 49 - 42 + 5 = 12$$

$$k = f(-1) = (-1)^2 - 6(-1) + 5 = 1 + 6 + 5 = 12$$

Method 2: Using the quadratic formula

Setting $f(x) = k$ gives the quadratic equation $x^2 - 6x + (5 - k) = 0$.

Using the quadratic formula, the solutions are:

$$x = \frac{6 \pm \sqrt{(-6)^2 - 4(1)(5-k)}}{2} = 3 \pm \sqrt{9 - (5-k)} = 3 \pm \sqrt{4 + k}$$

The distance between these two roots is the difference:

$$(3 + \sqrt{4+k}) - (3 - \sqrt{4+k}) = 2\sqrt{4+k}$$

Setting this distance equal to 8:

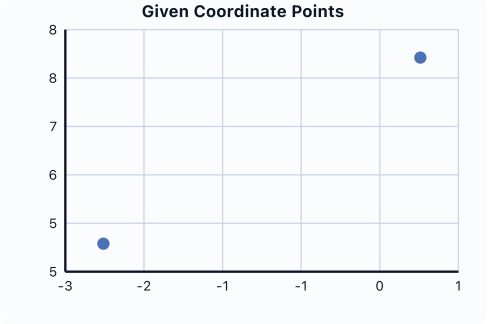
$$2\sqrt{4+k} = 8 \implies \sqrt{4+k} = 4 \implies 4 + k = 16 \implies k = 12$$

Common misconceptions:

- Choosing 8 confuses the horizontal distance between the points with the vertical position k.
- Choosing 4 results from solving $\sqrt{4+k}=4$ by incorrectly writing $4+k=8$ instead of squaring 4.
- Choosing -3 results from substituting the half-distance $x=4$ directly into $f(x)$ instead of shifting from the axis of symmetry $x=3$.

5 MCQ

In the standard (x, y) coordinate plane, a circle has its center at $(-3, 5)$ and passes through the point $(1, 8)$. What is the radius of this circle?



A 25

B 5 ✓

C 7

D $\sqrt{13}$

WORKED SOLUTION & VERIFICATION

The radius r of a circle is the distance from its center to any point on the circle. Using the distance formula between the center $(-3, 5)$ and the point on the circle $(1, 8)$:

$$r=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$

$$r=\sqrt{(1-(-3))^2+(8-5)^2}=\sqrt{4^2+3^2}=\sqrt{16+9}=\sqrt{25}=5$$

Thus, the radius of the circle is 5.

Common misconceptions:

- Choice B (7) mistakenly adds the coordinate differences directly: $4+3=7$.
- Choice C $(\sqrt{13})$ makes a sign error in subtracting the x-coordinates, computing $(1-3)^2+(8-5)^2=4+9=13$.
- Choice D (25) confuses the radius r with r^2.

6 MCQ

In the standard (x, y) coordinate plane, line L is perpendicular to the line $3x + 4y = 15$ and passes through the point $(3, 1)$. Line L intersects the line $2x + y = 17$ at the point (p, q). What is the value of $p + q$?

A 5

B **11** ✓

C 6

D 8

WORKED SOLUTION & VERIFICATION

Step 1: Find the slope of the given line $3x + 4y = 15$.
Rewriting in slope-intercept form gives:

$$4y = -3x + 15 \implies y = -\frac{3}{4}x + \frac{15}{4}$$

Thus, the slope of this line is $m_1 = -\frac{3}{4}$.
Step 2: Determine the slope of line L.
Since line L is perpendicular to the given line, its slope m_L is the negative reciprocal of m_1:

$$m_L = -\frac{1}{-\frac{3}{4}} = \frac{4}{3}$$

Step 3: Write the equation of line L.
Line L passes through $(3, 1)$ with slope $\frac{4}{3}$:

$$y - 1 = \frac{4}{3}(x - 3) \implies y = \frac{4}{3}x - 3$$

Step 4: Find the point of intersection (p, q) with the line $2x + y = 17$.
Substitute $y = \frac{4}{3}x - 3$ into $2x + y = 17$:

$$2x + (\frac{4}{3}x - 3) = 17$$

$$\frac{10}{3}x - 3 = 17 \implies \frac{10}{3}x = 20 \implies x = 6$$

Thus, $p = 6$.
Step 5: Solve for q.
Substitute $p = 6$ into $2x + y = 17$:

$$2(6) + q = 17 \implies 12 + q = 17 \implies q = 5$$

Step 6: Compute $p + q$.

$$p + q = 6 + 5 = 11$$

Common Misconceptions:
- Reporting only $p = 6$ or $q = 5$ fails to answer the question, which asks for $p + q$.
- Using the parallel slope $-\frac{3}{4}$ instead of the perpendicular slope $\frac{4}{3}$ yields $p = 11$ and $q = -5$, giving $p + q = 6$.

7 MCQ

If $f(x) = 2x - 1$ and $g(x) = x^2 + 3$, what is the value of $g(f(2))$?

A 9	B 12 ✓
C 13	D 21

WORKED SOLUTION & VERIFICATION

To find the composite value $g(f(2))$, evaluate from the inside out:

1. Evaluate the inner function at $x = 2$:

$$f(2) = 2(2) - 1 = 4 - 1 = 3$$

2. Substitute this result into the outer function $g(x)$:

$$g(3) = 3^2 + 3 = 9 + 3 = 12$$

Common Misconceptions:

- 13 results from reversing the order of operations to evaluate $f(g(2))$: $g(2) = 2^2 + 3 = 7$, and $f(7) = 2(7) - 1 = 13$.
- 21 results from multiplying the two values instead of composing them: $f(2) \cdot g(2) = 3 \cdot 7 = 21$.
- 9 results from squaring $f(2)$ but forgetting the constant term $+3$ in $g(x)$.

8 MCQ

What is the smallest positive integer n such that $126n$ is the square of an integer?

A 7	B 14 ✓
C 21	D 42

WORKED SOLUTION & VERIFICATION

To find the smallest positive integer n such that $126n$ is a perfect square, first find the prime factorization of 126:

$$126 = 2 \times 63 = 2 \times 3^2 \times 7$$

For an integer to be a perfect square, every prime factor in its prime factorization must have an even exponent.
In the prime factorization of 126:

- The factor 2 has an exponent of 1 (odd).
- The factor 3 has an exponent of 2 (even).
- The factor 7 has an exponent of 1 (odd).

To make all exponents even with the smallest positive integer n, n must provide at least one factor of 2 and one factor of 7:

$$n = 2^1 \times 7^1 = 14$$

Then $126 \times 14 = 1764 = 42^2$, which is a perfect square.
Common misconceptions:

- Choosing 7 only pairs the factor 7, leaving 2 with an odd exponent ($126 \times 7 = 882$, not a square).
- Choosing 21 multiplies by 3×7, leaving 2 unpaired.
- Choosing 42 gives $126 \times 42 = 5292$, which is not a square and is greater than 14.

9 MCQ

For a positive integer k, the greatest common divisor of k and 24 is 6, and the least common multiple of k and 40 is 600. What is the value of k?

A 120

B 300

C **150** ✓

D 75

WORKED SOLUTION & VERIFICATION

We use prime factorizations to determine the prime powers dividing k.
First, factor the given constants:

$$24 = 2^3 \cdot 3^1, \quad 40 = 2^3 \cdot 5^1, \quad 600 = 2^3 \cdot 3^1 \cdot 5^2$$

From $\text{lcm}(k, 40) = 600$:
For each prime p, the exponent in the least common multiple is the maximum of the exponents in k and 40:
- For $p = 3$: $\max(v_3(k), 0) = 1 \implies v_3(k) = 1$.
- For $p = 5$: $\max(v_5(k), 1) = 2 \implies v_5(k) = 2$.
- For $p > 5$: $v_p(k) = 0$ since no larger prime divides 600.
- For $p = 2$: $\max(v_2(k), 3) = 3 \implies v_2(k) \in \{0, 1, 2, 3\}$.

Next, use $\gcd(k, 24) = 6 = 2^1 \cdot 3^1$:
- The power of 2 in $\gcd(k, 24)$ is $\min(v_2(k), 3) = 1$, which requires $v_2(k) = 1$.

Combining these exponents:

$$k = 2^1 \cdot 3^1 \cdot 5^2 = 2 \cdot 3 \cdot 25 = 150$$

Common traps:
- Setting $v_2(k) = 0$ yields 75, but $\gcd(75, 24) = 3 \neq 6$.
- Setting $v_2(k) = 2$ yields 300, but $\gcd(300, 24) = 12 \neq 6$.

10 MCQ

In trapezoid $ABCD$, base $\overline{AB}$ is parallel to base $\overline{CD}$, and $\angle DAB = \angle ADC = 90^\circ$. The length of $\overline{AB}$ is x, the length of $\overline{CD}$ is $2x + 1$, and the length of the non-parallel side $\overline{BC}$ is $2x$. If the height $AD = 8$, what is the area of trapezoid $ABCD$?

A 34

B 80

C 128

D 64 ✓

WORKED SOLUTION & VERIFICATION

To find the area of trapezoid $ABCD$, we first determine the value of the parameter x.

1. **Deconstruct the trapezoid:**

Draw an altitude from vertex B perpendicular to $\overline{CD}$, intersecting $\overline{CD}$ at point E. Because $\angle DAB = \angle ADC = 90^\circ$ and $\overline{BE} \perp \overline{CD}$, quadrilateral $ABED$ is a rectangle. Therefore:

$$BE = AD = 8 \quad \text{and} \quad DE = AB = x$$

2. **Express the sides of right triangle $\triangle BEC$:**

Point E lies on $\overline{CD}$, so the remaining segment EC has length:

$$EC = CD - DE = (2x + 1) - x = x + 1$$

In right triangle $\triangle BEC$, the legs are $BE = 8$ and $EC = x + 1$, and the hypotenuse is $BC = 2x$.

3. **Apply the Pythagorean theorem:**

$$BE^2 + EC^2 = BC^2$$

$$8^2 + (x + 1)^2 = (2x)^2$$

$$64 + x^2 + 2x + 1 = 4x^2$$

$$3x^2 - 2x - 65 = 0$$

Factoring the quadratic equation:

$$(3x + 13)(x - 5) = 0$$

Since lengths must be positive, $x = 5$ (the negative root $x = -\frac{13}{3}$ is discarded).

4. **Calculate the bases and area:**

- $AB = x = 5$
- $CD = 2(5) + 1 = 11$
- Height $h = AD = 8$

The area of trapezoid $ABCD$ is:

$$\text{Area} = \frac{AB + CD}{2} \times h = \frac{5 + 11}{2} \times 8 = 8 \times 8 = 64$$

Common Traps:

- **34:** Correctly solves for all dimensions ($AB = 5$, $BC = 10$, $CD = 11$, $AD = 8$) but calculates the perimeter $5 + 10 + 11 + 8 = 34$ instead of the area.
- **80:** Erroneously substitutes the slant side $BC = 10$ in place of the perpendicular height $AD = 8$: $\frac{5+11}{2} \times 10 = 80$.
- **128:** Forgets the factor of $\frac{1}{2}$ in the trapezoid area formula: $(5 + 11) \times 8 = 128$.

11 MCQ

In the standard (x, y) coordinate plane, the system of equations

$$\begin{cases} y = |x^2 - 4| \\ y = 2x + k \end{cases}$$

has exactly 3 distinct real solutions (x, y). What is the sum of all possible values of the real constant k?

A 5

B 9 ✓

C 11

D 4

WORKED SOLUTION & VERIFICATION

The system represents the intersections of the absolute-value quadratic graph $y = |x^2 - 4|$ and the family of parallel lines $y = 2x + k$ of slope 2.

The piecewise definition of $y = |x^2 - 4|$ is:

$$y = \begin{cases} x^2 - 4, & \text{if } |x| \geq 2 \\ 4 - x^2, & \text{if } |x| < 2 \end{cases}$$

To find when there are exactly 3 intersection points, analyze the critical transitions as k increases:

1. **Tangency to the inverted arch** $y = 4 - x^2$:

Setting the derivative equal to the slope of the line:

$$\frac{d}{dx}(4 - x^2) = -2x = 2 \implies x = -1$$

At $x = -1$, $y = 4 - (-1)^2 = 3$.

Substituting $(-1, 3)$ into the line equation:

$$3 = 2(-1) + k \implies k = 5$$

For $k = 5$, the line is tangent to the inverted portion at $x = -1$ (1 solution) and intersects the outer branches $x^2 - 4 = 2x + 5 \implies x^2 - 2x - 9 = 0 \implies x = 1 \pm \sqrt{10}$. Since $1 + \sqrt{10} > 2$ and $1 - \sqrt{10} < -2$, these give 2 additional real solutions, totaling exactly **3 distinct solutions**.

2. **Passing through the left vertex/cusp** $(-2, 0)$:

Substitute $(-2, 0)$ into the line equation:

$$0 = 2(-2) + k \implies k = 4$$

For $k = 4$, the line $y = 2x + 4$ intersects:

- The inverted arch $4 - x^2 = 2x + 4 \implies x^2 + 2x = 0 \implies x = 0$ and $x = -2$.
- The right outer branch $x^2 - 4 = 2x + 4 \implies x^2 - 2x - 8 = 0 \implies (x - 4)(x + 2) = 0 \implies x = 4$.

Thus, the solutions are $x \in \{-2, 0, 4\}$, which gives exactly **3 distinct solutions**.

3. **Behavior in other intervals of** k:

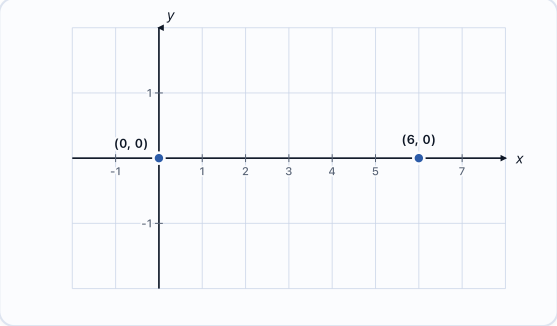
- For $k > 5$: 2 solutions (outer branches only).
- For $4 < k < 5$: 4 solutions (2 on the inverted arch, 2 on the outer branches).
- For $-4 < k < 4$: 2 solutions.
- For $k = -4$: 1 solution (passing through $(2, 0)$).
- For $k < -4$: 0 solutions.

Therefore, the only values of k that yield exactly 3 distinct solutions are $k = 4$ and $k = 5$.

Their sum is $4 + 5 = 9$.

12 MCQ

In the standard (x, y) coordinate plane, the vertices of $\triangle OAB$ are $O(0, 0)$, $A(6, 0)$, and $B(2, 4)$. A vertical line with equation $x = k$, where $0 < k < 6$, divides $\triangle OAB$ into two regions of equal area. What is the value of k?



A $6 - 2\sqrt{3}$ ✓

B $\sqrt{6}$

C $\frac{8}{3}$

D 3

WORKED SOLUTION & VERIFICATION

1. **Find the total area of $\triangle OAB$:**

Taking the side along the x-axis as the base, the length of base OA is $6 - 0 = 6$.

The height is the y-coordinate of vertex $B(2, 4)$, which is 4.

$$\text{Area}(\triangle OAB) = \frac{1}{2} \times 6 \times 4 = 12$$

To divide the triangle into two regions of equal area, each region must have an area of $\frac{12}{2} = 6$.

2. **Determine the domain of k:**

- If $0 < k \leq 2$, the vertical line $x = k$ intersects the side OB, which has equation $y = 2x$. The region to the left is a triangle with base k and height $2k$, giving an area of:

$$\text{Area} = \frac{1}{2} \times k \times 2k = k^2$$

Setting $k^2 = 6$ yields $k = \sqrt{6} \approx 2.45$. However, this contradicts $k \leq 2$. Therefore, k must satisfy $2 < k < 6$.

3. **Calculate the area for $2 < k < 6$:**

For $k > 2$, the line $x = k$ lies to the right of vertex $B(2, 4)$. The region to the right of $x = k$ is a triangle formed by the line $x = k$, the segment AB, and the x-axis.

- Find the equation of the line through $B(2, 4)$ and $A(6, 0)$:

$$\text{Slope } m = \frac{0 - 4}{6 - 2} = -1$$

$$y - 0 = -(x - 6) \implies y = 6 - x$$

- At $x = k$, the height of this right-triangular region is $y = 6 - k$.
- The base along the x-axis runs from $x = k$ to $x = 6$, so its length is $6 - k$.
- The area of this right triangle is:

$$\text{Area} = \frac{1}{2}(6 - k)(6 - k) = \frac{1}{2}(6 - k)^2$$

4. **Solve for k:**

$$\frac{1}{2}(6 - k)^2 = 6 \implies (6 - k)^2 = 12$$

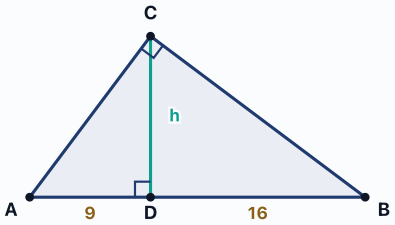
Since $k < 6$, $6 - k > 0$:

$$6 - k = \sqrt{12} = 2\sqrt{3} \implies k = 6 - 2\sqrt{3}$$

Common Misconceptions:
- Setting the left region area $k^2 = 6$ gives $\sqrt{6}$, but this fails because the boundary $x = k$ is past the vertex at $x = 2$.
- Choosing $\frac{8}{3}$ mistakenly assumes that any vertical line passing through the centroid $x_c = \frac{0+6+2}{3} = \frac{8}{3}$ bisects the area of a triangle, which is only true if the line passes through a vertex.

13 MCQ

In the figure below, $\triangle ABC$ is a right triangle with hypotenuse $\overline{AB}$, and altitude $\overline{CD}$ of length h is drawn to $\overline{AB}$. The altitude divides the hypotenuse into two segments of lengths $AD = 9$ and $DB = 16$. A circle with center O_1 is inscribed in $\triangle ACD$, and a circle with center O_2 is inscribed in $\triangle BCD$. What is the distance between the centers O_1 and O_2?



A 5

B 7

C $7\sqrt{2}$

D $5\sqrt{2}$ ✓

WORKED SOLUTION & VERIFICATION

We solve the problem in five systematic steps:

1. **Determine altitude CD**:

By the Geometric Mean Theorem for right triangles, the altitude to the hypotenuse satisfies:

$$CD^2 = AD \cdot DB = 9 \cdot 16 = 144 \implies CD = 12.$$

2. **Find the side lengths of $\triangle ACD$ and $\triangle BCD$**:

- In right triangle $\triangle ACD$, the legs are $AD = 9$ and $CD = 12$. By the Pythagorean theorem:

$$AC = \sqrt{9^2 + 12^2} = \sqrt{81 + 144} = \sqrt{225} = 15.$$

- In right triangle $\triangle BCD$, the legs are $DB = 16$ and $CD = 12$. By the Pythagorean theorem:

$$BC = \sqrt{16^2 + 12^2} = \sqrt{256 + 144} = \sqrt{400} = 20.$$

3. **Compute the inradius of each triangle**:

For any right triangle with legs a, b and hypotenuse c, the inradius is given by $r = \frac{a+b-c}{2}$:

- For $\triangle ACD$:

$$r_1 = \frac{9 + 12 - 15}{2} = \frac{6}{2} = 3.$$

- For $\triangle BCD$:

$$r_2 = \frac{16 + 12 - 20}{2} = \frac{8}{2} = 4.$$

4. **Set up a coordinate system to locate incenters O_1 and O_2**:

Let point D be the origin $(0, 0)$, with hypotenuse $\overline{AB}$ along the x-axis and altitude $\overline{CD}$ along the positive y-axis.

- $\triangle ACD$ lies in the second quadrant with legs along the negative x-axis and positive y-axis. Since the incircle has radius $r_1 = 3$ and is tangent to both axes, its center is:

$$O_1 = (-3, 3).$$

- $\triangle BCD$ lies in the first quadrant with legs along the positive x-axis and positive y-axis. Since the incircle has radius $r_2 = 4$ and is tangent to both axes, its center is:

$$O_2 = (4, 4).$$

5. **Calculate the distance between O_1 and O_2:**
Using the distance formula:

$$O_1O_2 = \sqrt{(4-(-3))^2 + (4-3)^2} = \sqrt{7^2 + 1^2} = \sqrt{49+1} = \sqrt{50} = 5\sqrt{2}.$$

Misconception Analysis:
- 5 results from confusing the distance between incenters with the inradius r of the overall triangle $\triangle ABC$, where $r = \sqrt{r_1^2 + r_2^2} = \sqrt{3^2 + 4^2} = 5$.
- 7 results from simply adding the two inradii ($3 + 4 = 7$), failing to account for the two-dimensional distance.
- $7\sqrt{2}$ results from incorrectly taking the vertical distance as $r_1 + r_2 = 7$ instead of $|r_2 - r_1| = 1$.

14 MCQ

Consider the system of linear equations below:

$$\begin{cases} 3x - 2y = 8 \\ kx + 6y = 5 \end{cases}$$

For which of the following values of the constant k does the system have no solution (x, y)?

A 4

B 9

C -4

D **-9** ✓

WORKED SOLUTION & VERIFICATION

A system of two linear equations in two variables has no solution if and only if the lines represented by the equations are parallel and distinct (they have the same slope but different y-intercepts).

First, rewrite each equation in slope-intercept form $y = mx + b$:

1. For the first equation:

$$3x - 2y = 8 \implies -2y = -3x + 8 \implies y = \frac{3}{2}x - 4$$

The slope of the first line is $m_1 = \frac{3}{2}$ and the y-intercept is -4.

2. For the second equation:

$$kx + 6y = 5 \implies 6y = -kx + 5 \implies y = -\frac{k}{6}x + \frac{5}{6}$$

The slope of the second line is $m_2 = -\frac{k}{6}$ and the y-intercept is $\frac{5}{6}$.

Since the y-intercepts -4 and $\frac{5}{6}$ are unequal, the lines will be distinct and parallel whenever their slopes are equal:

$$\frac{3}{2} = -\frac{k}{6}$$

Cross-multiplying yields:

$$2(-k) = 3(6) \implies -2k = 18 \implies k = -9$$

Alternatively, using the coefficient determinant criterion for parallelism:

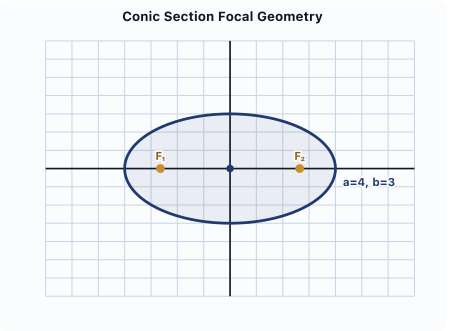
$$\det\begin{pmatrix} 3 & -2 \\ k & 6 \end{pmatrix} = 3(6) - (-2)(k) = 18 + 2k = 0 \implies k = -9$$

Common Misconceptions:

- Choosing 9: Results from a sign error when isolating y or evaluating the determinant $18 - 2k = 0$.
- Choosing -4 or 4: Results from incorrectly combining coefficients column-wise (e.g., $3k + 12 = 0$) rather than cross-multiplying.

15 MCQ

In the standard (x, y) coordinate plane, what is the length of the major axis of the ellipse defined by the equation $\frac{(x-4)^2}{49} + \frac{(y+2)^2}{16} = 1$?



A 7

B 8

C 14 ✓

D 49

WORKED SOLUTION & VERIFICATION

The standard form of a horizontal ellipse is $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$, where $a > b > 0$.

1. Identify the values of a^2 and b^2 from the given equation:

$$a^2 = 49 \quad \text{and} \quad b^2 = 16$$

2. Solve for the semi-major axis length a:

$$a = \sqrt{49} = 7$$

3. The total length of the major axis is $2a$:

$$2a = 2 \times 7 = 14$$

Common misconceptions:

- Choosing 7 confuses the semi-major axis length a with the full major axis length $2a$.
- Choosing 8 calculates the length of the minor axis, $2b = 2\sqrt{16} = 8$.
- Choosing 49 uses the denominator a^2 directly without taking the square root.

16 MCQ

Each interior angle of a regular polygon measures 144°. How many sides does the polygon have?

A 8

B **10** ✓

C 9

D 12

WORKED SOLUTION & VERIFICATION

To find the number of sides n of a regular polygon from its interior angle measure, we can use the relationship between the interior angle and the exterior angle.

Each exterior angle of a regular polygon is supplementary to its corresponding interior angle:

$$\text{Exterior Angle} = 180^\circ - 144^\circ = 36^\circ$$

The sum of the exterior angles of any convex polygon is always 360°. Since all n exterior angles in a regular polygon are equal:

$$n = \frac{360^\circ}{36^\circ} = 10$$

Alternatively, using the formula for the interior angle of a regular n-gon directly:

$$\frac{(n-2) \times 180^\circ}{n} = 144^\circ$$

Multiplying both sides by n:

$$180n - 360 = 144n$$

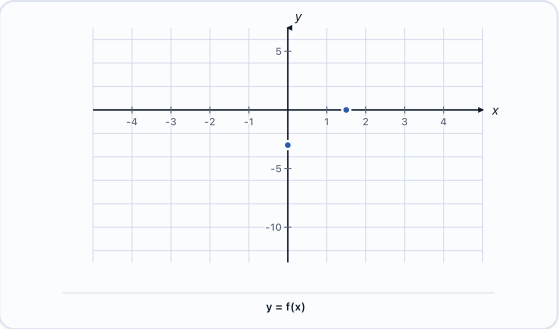
$$36n = 360 \implies n = 10$$

Thus, the polygon has 10 sides.

- Choice A (8) corresponds to an interior angle of $\frac{(8-2)\times 180^\circ}{8} = 135^\circ$.
- Choice B (9) corresponds to an interior angle of $\frac{(9-2)\times 180^\circ}{9} = 140^\circ$.
- Choice D (12) corresponds to an interior angle of $\frac{(12-2)\times 180^\circ}{12} = 150^\circ$.

17 MCQ

In the standard (x, y) coordinate plane, the line $y = 2x - 3$ intersects the parabola $y = x^2 - 6x + c$ at two distinct points, P and Q. If the distance between P and Q is $4\sqrt{5}$, what is the value of the constant c?



A -7

B 12

C 15

D 9 ✓

WORKED SOLUTION & VERIFICATION

To find the points of intersection, set the two equations equal:

$$x^2 - 6x + c = 2x - 3$$

Rearranging into standard quadratic form gives:

$$x^2 - 8x + (c + 3) = 0$$

Let the roots of this equation be x_1 and x_2. By Vieta's formulas, the sum and product of the roots are:

$$x_1 + x_2 = 8 \quad \text{and} \quad x_1 x_2 = c + 3$$

The horizontal distance between the intersection points is given by $|x_2 - x_1|$. We can calculate $(x_2 - x_1)^2$ using the identity:

$$(x_2 - x_1)^2 = (x_1 + x_2)^2 - 4x_1 x_2 = 8^2 - 4(c + 3) = 64 - 4c - 12 = 52 - 4c$$

Since both points lie on the line $y = 2x - 3$, their vertical difference is:

$$y_2 - y_1 = (2x_2 - 3) - (2x_1 - 3) = 2(x_2 - x_1)$$

Using the distance formula, the distance between P and Q is:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} = \sqrt{(x_2 - x_1)^2 + 4(x_2 - x_1)^2} = \sqrt{5(x_2 - x_1)^2} = \sqrt{5}|x_2 - x_1|$$

We are given $d = 4\sqrt{5}$, so:

$$\sqrt{5}|x_2 - x_1| = 4\sqrt{5} \implies |x_2 - x_1| = 4$$

Squaring both sides:

$$(x_2 - x_1)^2 = 16$$

Substitute this back into our expression for $(x_2 - x_1)^2$:

$$52 - 4c = 16$$

$$4c = 36 \implies c = 9$$

Common Errors:

- Forgetting that the distance formula incorporates the line's slope $m = 2$ (i.e., setting $|x_2 - x_1| = 4\sqrt{5}$) leads to $52 - 4c = 80 \implies c = -7$.
- Forgetting to square the difference 4 (i.e., setting $52 - 4c = 4$) leads to $c = 12$.
- Making a sign error when rearranging -3 into $c - 3$ gives $76 - 4c = 16 \implies c = 15$.

18 MCQ

In the standard (x, y) coordinate plane, a right triangle has vertices at $A(-3, -1)$, $B(5, 5)$, and $C(4, k)$, where the right angle is at vertex C and $k > 0$. What is the value of k?

A 8

B 2

C 4

D 6 ✓

WORKED SOLUTION & VERIFICATION

To find the unknown parameter k, use the condition that $\angle ACB = 90^\circ$, which means line segment $\overline{AC}$ is perpendicular to line segment $\overline{BC}$.

1. Find the slope of $\overline{AC}$:

$$m_{AC} = \frac{k - (-1)}{4 - (-3)} = \frac{k + 1}{7}$$

2. Find the slope of $\overline{BC}$:

$$m_{BC} = \frac{k - 5}{4 - 5} = \frac{k - 5}{-1} = -(k - 5)$$

3. Since perpendicular lines have slopes that multiply to -1:

$$m_{AC} \cdot m_{BC} = -1$$

$$\left(\frac{k + 1}{7}\right)(-(k - 5)) = -1$$

$$(k + 1)(k - 5) = 7$$

$$k^2 - 4k - 5 = 7$$

$$k^2 - 4k - 12 = 0$$

4. Factor the quadratic equation:

$$(k - 6)(k + 2) = 0$$

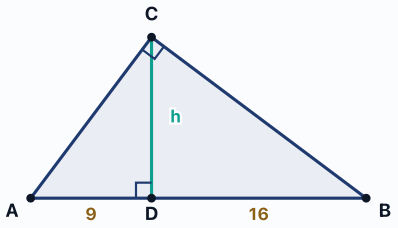
This gives $k = 6$ or $k = -2$.

Since the problem specifies that $k > 0$, we reject $k = -2$ and obtain $k = 6$.

Common Trap: Solving $(k - 2)^2 = 16$ (from the circumcircle centered at $(1, 2)$) and forgetting to add 2 leads to the misconception $k = 4$.

19 MCQ

In the figure below, $\triangle ABC$ is a right triangle with right angle at B, and $\overline{BD}$ is an altitude to hypotenuse $\overline{AC}$ with length h. If $AD = 9$ and $DC = 16$, what is the perimeter of $\triangle ABC$?



A 47

B 60 ✓

C 50

D 72

WORKED SOLUTION & VERIFICATION

By the Geometric Mean (Altitude) Theorem in right triangle ABC, the altitude to the hypotenuse satisfies:

$$h^2 = AD \cdot DC$$

Given $AD = 9$ and $DC = 16$:

$$h^2 = 9 \cdot 16 = 144 \implies h = 12$$

Next, use the Geometric Mean (Leg) Theorem or the Pythagorean theorem in the smaller right triangles to find the lengths of legs $\overline{AB}$ and $\overline{BC}$:

$$AB^2 = AD \cdot AC = 9 \cdot (9 + 16) = 9 \cdot 25 = 225 \implies AB = 15$$

$$BC^2 = DC \cdot AC = 16 \cdot (9 + 16) = 16 \cdot 25 = 400 \implies BC = 20$$

The length of hypotenuse $\overline{AC}$ is $AD + DC = 9 + 16 = 25$.
Therefore, the perimeter of $\triangle ABC$ is:

$$\text{Perimeter} = AB + BC + AC = 15 + 20 + 25 = 60$$

Common error: Adding the altitude instead of the hypotenuse yields $15 + 20 + 12 = 47$, or adding all internal and external segments yields $15 + 20 + 25 + 12 = 72$.

20 MCQ

If the real numbers x and y satisfy the system of equations

$$x^3 - 3xy^2 = 10$$
$$3x^2y - y^3 = 5$$

what is the value of $x^2 + y^2$?

A 25

B $\sqrt{5}$

C $5\sqrt{5}$

D 5 ✓

WORKED SOLUTION & VERIFICATION

We can solve this system using algebraic expansion or complex numbers.

Method 1: Squaring and Adding

Square both equations and add them together:

$$(x^3 - 3xy^2)^2 + (3x^2y - y^3)^2 = 10^2 + 5^2 = 100 + 25 = 125$$

Expanding the left-hand side:

$$(x^6 - 6x^4y^2 + 9x^2y^4) + (9x^4y^2 - 6x^2y^4 + y^6) = x^6 + 3x^4y^2 + 3x^2y^4 + y^6$$

Notice that this expression is the binomial expansion of $(x^2 + y^2)^3$:

$$(x^2 + y^2)^3 = 125$$

Taking the real cube root of both sides gives:

$$x^2 + y^2 = \sqrt[3]{125} = 5$$

Method 2: Complex Numbers

Let $z = x + iy$. By De Moivre's formula / binomial expansion:

$$z^3 = (x + iy)^3 = (x^3 - 3xy^2) + i(3x^2y - y^3) = 10 + 5i$$

Taking the modulus of both sides:

$$|z^3| = |z|^3 = |10 + 5i| = \sqrt{10^2 + 5^2} = \sqrt{125} = 5\sqrt{5}$$

Since $|z| = \sqrt{x^2 + y^2}$, we have:

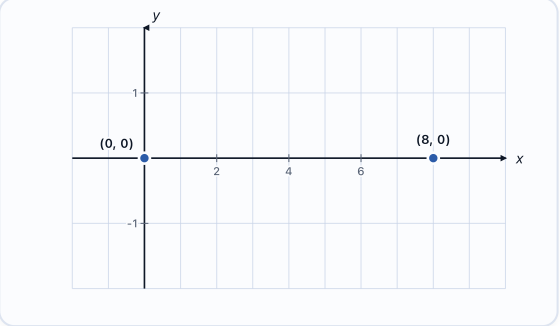
$$(x^2 + y^2)^{3/2} = 5^{3/2} \implies x^2 + y^2 = 5$$

Misconception Traps:

- Choice C, $5\sqrt{5}$, occurs if one incorrectly assumes the sum of squares yields $(x^2 + y^2)^2$ instead of $(x^2 + y^2)^3$, computing $\sqrt{125}$.
- Choice D, 25, arises from mistakenly dividing 125 by 5 rather than taking the cube root.
- Choice A, $\sqrt{5}$, results from erroneously taking the fourth root or an extra square root.

21 MCQ

In the standard (x, y) coordinate plane, a triangle has vertices at $A(0, 0)$, $B(8, 0)$, and $C(2, 6)$. A vertical line with equation $x = k$, where $0 < k < 8$, divides the triangle into two regions of equal area. What is the value of k?



A 4

B $\frac{10}{3}$

C $2\sqrt{6}$

D $8 - 2\sqrt{6}$ ✓

WORKED SOLUTION & VERIFICATION

1. **Find the total area of $\triangle ABC$:**

The base along the x-axis connects $A(0, 0)$ and $B(8, 0)$, so the base length is $b = 8$.

The vertex $C(2, 6)$ gives a height of $h = 6$.

$$\text{Area}(\triangle ABC) = \frac{1}{2} \times 8 \times 6 = 24$$

Each of the two regions must therefore have an area of $\frac{24}{2} = 12$.

2. **Determine the location of $x = k$:**

The vertical line through C is $x = 2$. The portion of $\triangle ABC$ to the left of $x = 2$ is a triangle with base on the x-axis from $x = 0$ to $x = 2$ (length 2) and height 6:

$$\text{Area}_{x \le 2} = \frac{1}{2} \times 2 \times 6 = 6$$

Since $6 < 12$, the vertical line dividing the triangle in half must lie to the right of $x = 2$, so $2 < k < 8$.

3. **Set up the area of the region to the right of $x = k$:**

For $2 < k < 8$, the region to the right of $x = k$ is a triangle bounded by:

- The vertical line $x = k$,
- The horizontal x-axis (from $x = k$ to $x = 8$),
- The line segment BC.

The equation of the line passing through $C(2, 6)$ and $B(8, 0)$ has slope:

$$m = \frac{0 - 6}{8 - 2} = -1$$

Thus, the equation of line BC is $y - 0 = -(x - 8)$, or $y = 8 - x$.

At $x = k$, the y-coordinate on BC is $8 - k$. Since the vertical line is perpendicular to the x-axis, the region to the right is a right triangle with legs of length $8 - k$ (horizontal) and $8 - k$ (vertical).

4. **Solve for k:**

$$\text{Area} = \frac{1}{2}(8 - k)^2 = 12$$

$$(8 - k)^2 = 24 \implies 8 - k = \sqrt{24} = 2\sqrt{6}$$

$$k = 8 - 2\sqrt{6}$$

Common Misconceptions:
- Choosing 4 assumes the vertical line bisects the base to bisect area, which only holds for symmetric triangles or lines through a vertex.
- Choosing $\frac{10}{3}$ uses the x-coordinate of the centroid, but a vertical line through the centroid does not generally bisect the area of a triangle.
- Choosing $2\sqrt{6}$ results from solving $8 - k = 2\sqrt{6}$ but mistakenly dropping the constant 8.

22 MCQ

In $\triangle ABC$, the side lengths are $AB = 9$, $AC = 8$, and $BC = 7$. A line segment $\overline{DE}$, with D lying on $\overline{AB}$ and E lying on $\overline{AC}$, divides $\triangle ABC$ into two regions that have equal areas and equal perimeters. What is the length of $\overline{DE}$?

A $\frac{7\sqrt{2}}{2}$

B $4\sqrt{3}$

C $2\sqrt{6}$ ✓

D $6\sqrt{2}$

WORKED SOLUTION & VERIFICATION

The line segment $\overline{DE}$ divides $\triangle ABC$ into two regions: $\triangle ADE$ and quadrilateral $BDEC$.

1. **Equal Perimeters:**

The perimeter of $\triangle ADE$ is $AD + AE + DE$.

The perimeter of quadrilateral $BDEC$ is $DB + BC + CE + DE = (AB - AD) + BC + (AC - AE) + DE$.

Equating the two perimeters and subtracting DE from both sides gives:

$$AD + AE = (AB + BC + AC) - (AD + AE)$$

$$2(AD + AE) = 9 + 7 + 8 = 24 \implies AD + AE = 12$$

2. **Equal Areas:**

Using the sine formula for area, $\text{Area}(\triangle ABC) = \frac{1}{2}AB \cdot AC \sin A$ and $\text{Area}(\triangle ADE) = \frac{1}{2}AD \cdot AE \sin A$.

Since $\text{Area}(\triangle ADE) = \frac{1}{2}\text{Area}(\triangle ABC)$:

$$AD \cdot AE = \frac{1}{2}(AB \cdot AC) = \frac{1}{2}(9 \times 8) = 36$$

3. **Finding AD and AE:**

The lengths AD and AE satisfy the system $AD + AE = 12$ and $AD \cdot AE = 36$, which are the roots of:

$$t^2 - 12t + 36 = (t - 6)^2 = 0 \implies AD = 6 \text{ and } AE = 6$$

Since $6 < 9$ and $6 < 8$, both points lie strictly within segments $\overline{AB}$ and $\overline{AC}$.

4. **Finding DE via Law of Cosines:**

First, find $\cos A$ in $\triangle ABC$:

$$\cos A = \frac{AB^2 + AC^2 - BC^2}{2(AB)(AC)} = \frac{9^2 + 8^2 - 7^2}{2(9)(8)} = \frac{81 + 64 - 49}{144} = \frac{96}{144} = \frac{2}{3}$$

Now apply the Law of Cosines in $\triangle ADE$:

$$DE^2 = AD^2 + AE^2 - 2(AD)(AE)\cos A = 6^2 + 6^2 - 2(6)(6)\left(\frac{2}{3}\right) = 72 - 48 = 24$$

$$DE = \sqrt{24} = 2\sqrt{6}$$

Common misconceptions:

- Incorrectly assuming $\overline{DE} \parallel \overline{BC}$ yields $DE = \frac{BC}{\sqrt{2}} = \frac{7\sqrt{2}}{2}$.
- Assuming $\angle A = 90^\circ$ yields $\sqrt{6^2 + 6^2} = 6\sqrt{2}$.

23 MCQ

A committee of 4 people is to be selected at random from a group of 10 candidates consisting of 4 mathematicians, 3 physicists, and 3 engineers. Given that the selected committee contains at least one person from each of the three professions, what is the probability that exactly 2 mathematicians are on the committee?

- **A** $\frac{3}{7}$ ✓
- **B** $\frac{1}{3}$
- **C** $\frac{9}{35}$
- **D** $\frac{3}{5}$

WORKED SOLUTION & VERIFICATION

To find the conditional probability, we determine the number of committee selections that satisfy the given condition (the reduced sample space) and the number of those selections that contain exactly 2 mathematicians.

1. **Analyze the condition:**

The committee must have 4 members chosen from 3 professions with at least one member from each profession. By the Pigeonhole Principle, the members must be partitioned as $2+1+1$. Thus, exactly one profession will have 2 representatives, and the remaining two professions will each have 1 representative.

2. **Count the outcomes for each mutually exclusive composition:**

- **Case 1: 2 mathematicians, 1 physicist, 1 engineer**

$$\binom{4}{2} \times \binom{3}{1} \times \binom{3}{1} = 6 \times 3 \times 3 = 54$$

- **Case 2: 1 mathematician, 2 physicists, 1 engineer**

$$\binom{4}{1} \times \binom{3}{2} \times \binom{3}{1} = 4 \times 3 \times 3 = 36$$

- **Case 3: 1 mathematician, 1 physicist, 2 engineers**

$$\binom{4}{1} \times \binom{3}{1} \times \binom{3}{2} = 4 \times 3 \times 3 = 36$$

3. **Calculate the total number of outcomes satisfying the condition:**

$$N(\text{condition}) = 54 + 36 + 36 = 126$$

4. **Compute the conditional probability:**

The number of favorable outcomes (where exactly 2 mathematicians are chosen) is 54.

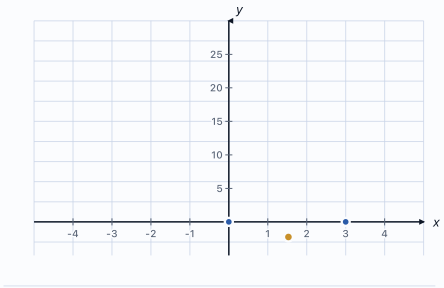
$$P(\text{2 mathematicians} \mid \text{at least 1 of each}) = \frac{54}{126} = \frac{3}{7}$$

Common Misconceptions:

- $\frac{1}{3}$ assumes all three professions are equally likely to have 2 members, failing to account for the larger pool of mathematicians (4 candidates vs. 3).
- $\frac{9}{35}$ is the unconditional probability $\frac{54}{\binom{10}{4}} = \frac{54}{210}$, forgetting to restrict the sample space to the given condition.
- $\frac{3}{5}$ represents the overall probability of the condition occurring, $\frac{126}{210}$, rather than the required conditional probability.

24 MCQ

In the standard (x, y) coordinate plane, at what values of x do the graphs of $y = x^2 - 3x$ and $y = 2x - 4$ intersect?



A $x = -2$ and $x = 2$

B $x = 2$ and $x = 3$

C $x = -4$ and $x = -1$

D $x = 1$ **and** $x = 4$ ✓

WORKED SOLUTION & VERIFICATION

To find the x-values of the intersection points, set the two expressions for y equal to each other:

$$x^2 - 3x = 2x - 4$$

Subtract $2x$ and add 4 to both sides to write the quadratic equation in standard form:

$$x^2 - 5x + 4 = 0$$

Factor the quadratic expression:

$$(x - 1)(x - 4) = 0$$

Setting each factor equal to zero gives:

$$x - 1 = 0 \implies x = 1$$

$$x - 4 = 0 \implies x = 4$$

Thus, the graphs intersect at $x = 1$ and $x = 4$.

Misconception notes:

- Choosing $x = -4$ and $x = -1$ typically results from sign errors when moving terms across the equals sign, leading to $x^2 + 5x + 4 = 0$.
- Choosing $x = 2$ and $x = 3$ results from mistakenly factoring with terms whose product is 6 rather than 4.

25 MCQ

In the standard (x, y) coordinate plane, what are the center and the radius of the circle with equation $(x - 4)^2 + (y + 6)^2 = 49$?

- **A** **Center $(4, -6)$, radius 7** ✓
- **B** Center $(-4, 6)$, radius 7
- **C** Center $(4, -6)$, radius 49
- **D** Center $(-4, 6)$, radius 49

WORKED SOLUTION & VERIFICATION

The standard form of the equation of a circle in the (x, y) coordinate plane is:

$$(x - h)^2 + (y - k)^2 = r^2$$

where (h, k) is the center of the circle and r is the radius (with $r > 0$).

Comparing the given equation $(x - 4)^2 + (y + 6)^2 = 49$ to the standard form:

1. For the x-coordinate of the center: $x - h = x - 4 \implies h = 4$.
2. For the y-coordinate of the center: $y - k = y + 6 = y - (-6) \implies k = -6$.

Therefore, the center is $(4, -6)$.

3. For the radius: $r^2 = 49 \implies r = \sqrt{49} = 7$.

Thus, the center is $(4, -6)$ and the radius is 7.

Common misconceptions:

- Reversing the signs of h and k leads to the incorrect center $(-4, 6)$.
- Forgetting to take the square root of the constant on the right side leads to the incorrect radius 49.